AYUSH SHARMA

Software Development Engineer II | Distributed Systems | Fintech, Risk & Lending Platforms

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PROFESSIONAL SUMMARY

Software engineer with **3+ years** building distributed **Java and Spring Boot** systems for fraud detection, credit underwriting and BNPL lending at Jupiter Money. Own **12+ production services** that process **1M events a day** at **99.9% uptime**. Shipped work that drove **30% revenue growth**, lifted loan approvals by **15%+**, cut KYC cost by **~25%** and took lending partners live at **₹35 Cr+ a month** in disbursals. Also built an AI on-call agent using **MCP** that cut alert triage effort by **95%**.

TECHNICAL SKILLS

Languages:	**Java**, **Python**, Scala, Kotlin, JavaScript, **SQL**
Backend:	**Distributed Systems**, **Microservices**, **Spring Boot**, **REST APIs**, Event-Driven Architecture, Rules / Policy Engines
AI & GenAI:	**Model Context Protocol (MCP)**, **Agentic AI workflows**, Prompt engineering, AI-assisted coding (Claude)
Data & Streaming:	**Apache Kafka**, Akka Streams, **Apache Airflow**, **Databricks**
Databases:	**PostgreSQL**, **DynamoDB**, **Redis**, **Elasticsearch**, TimescaleDB
Cloud & DevOps:	**AWS** (Lambda, EC2, S3), **Docker**, **Kubernetes**, Spinnaker, **CI/CD**
Observability:	**Prometheus**, **Grafana**, Kibana, Alerting, Incident Response, A/B Experimentation

PROFESSIONAL EXPERIENCE

Software Development Engineer II
October 2026 – Present

Jupiter Money | Loans Tech | Bangalore, India

- Built a **bank statement income fallback** for applicants whose Account Aggregator consent fails, so they get an income check instead of a decline, **lifting approvals by 15%+**.
- **Cut KYC cost by ~25%** by replacing video KYC with liveness checks for loans under ₹1L and moving DigiLocker traffic to cheaper CKYC, while keeping stricter checks for high-risk users.
- Turned the first partner integration into a reusable template, **cutting partner go-live time from 8+ weeks to about 2**, and took **KSF co-lending** live at **₹35 Cr a month** in disbursals.
- Integrated separate **ML models for new and repeat customers** into live credit decisioning, **reducing default rates by 10%+**.
- Migrated co-lending hard pulls from Equifax to **Experian** in a staged rollout (2%, 10%, then 100%) with **no impact on live loans**.
- Fixed a **500GB+ Elasticsearch index** for real-time fraud decisions that had stopped indexing new data, and added shard management and data expiry so it can't recur.

Software Development Engineer I
June 2024 – October 2026

Jupiter Money | Risk Tech | Bangalore, India

- Owned **12+ production services** processing **1M events a day** across fraud, underwriting and BNPL lending at **99.9% uptime**.
- Built the **SuperMoney BNPL credit line** and credit limit engine from scratch, which drove **30% revenue growth** for the lending business.
- Took **4 lending partners** live end to end (SuperMoney, GoKwik, ShopSe and KSF), with **₹35 Cr+ in combined monthly disbursals**.
- Integrated **6+ third-party fraud and risk signals** (device, identity, telecom, bureau and government data) into onboarding and pre-disbursal checks, **blocking 30% of fraudulent users** at onboarding.
- Shipped **~40 credit policy changes** and new ML underwriting models, **cutting false positives by 20%** and lifting true negatives by 10%.
- Built an automated **exposure circuit breaker** in the credit decision path that has **triggered 20+ times** in production, and the team's **A/B testing framework** for credit policies (10+ experiments).
- Moved Airflow DAGs to Databricks and caching to Redis Cloud, and upgraded Java 17 to 21, **cutting infrastructure cost by 30%**.
- Built an **MCP-based AI on-call agent** (Claude, Grafana, Slack, GitHub, Jira) that runs on every alert and **cuts triage effort by up to 95%**.

Software Development Engineering Intern
November 2023 – June 2024

Jupiter Money | Risk Tech | Bangalore, India

- Built real-time ingestion pipelines with **Kafka** and **Akka Streams** into DynamoDB and PostgreSQL, with **latency under 100ms**.
- Shipped Airflow DAGs, AWS Lambdas and Spring Boot APIs, and **cut deployment time from 30 minutes to 5**.

KEY PROJECTS

Movie Ticket Booking System [GitHub]
Java 21, Spring Boot 3, PostgreSQL, Flyway, Testcontainers

- Handles concurrent seat bookings with a **conditional UPDATE in PostgreSQL instead of locks**. In a **100-thread race test**, exactly one booking wins and no seat is double-booked.
- Runs payment calls outside DB transactions with **idempotency keys** and a **transactional outbox**; all **85 tests** pass on real Postgres.

Metrics Publisher Service [GitHub]
Java 17, Spring Boot, Apache Kafka, TimescaleDB, Redis, Docker

- Time-series metrics system on a **4-stage Kafka pipeline**, with **Redis cardinality limits** (1K series per metric) and **T-Digest** quantiles.
- REST API for P95, SUM, AVG and RATE queries over time buckets, with 90-day retention and compression.

EDUCATION

B.Tech, Computer Science and Engineering
2020 – 2024

Indraprastha Institute of Information Technology, Delhi (IIITD) | CGPA: 8.0 / 10

CERTIFICATIONS, RECOGNITION & ADDITIONAL

- **Best Performer of the Quarter**, Jupiter Money (for consistent delivery and production ownership)
- **ARTPARK Fellowship**, IISc Bangalore (for research contributions)
- **Amazon ML Summer School**, Amazon, 2022 (competitively selected)
- **Teaching Assistant**, Basic Electronics at IIITD (300+ undergraduates)